Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Mason Pust	mhpust@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My project will be a Construction Project Manager. It is very relevant to my civil engineering major, and will be helpful to keep track of ongoing projects, materials, progress, and future plans.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are several websites/apps that do similar things, but they all seem to have paywalls.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would want this application to be used mainly by people who oversee construction or other similar things, and have to keep track of progress made on different projects, as well as scheduling and budgeting issues.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal for the program is to be able to have different buttons that return different things, like finished projects, ongoing projects, costs, and other similar things.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to provide things like Project name, percent complete, and project location and possibly things like worker names and costs.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer can filter projects based on their completion percentage or budget, it can calculate the average cost of projects, and other things of that nature. Another thing that would be useful is if it can calculate which project is highest priority to complete based on certain criteria.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a recommendation on which project to focus on, a schedule, or budget information depending on the buttons they click and input they provide.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Project Completion, Project Name, Site Location, Budget, Cost of Project, Workers, Time Spent, Materials, Project Type,

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I will definitely need include Project Completion, Project Name, Cost of Project, Materials, Project Type
Details I can likely ignore include Workers, Site Location

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The part that seems the most difficult is organizing the dataclasses and code neatly, as well as determining what project is most important based on the criteria that I will figure out.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

If I do end up using an LLM to help code, what I would do is try the code myself, and then if I end up completely stuck, I will ask for help on the next step, but not for the LLM to code it for me.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to have a functioning website that could actually be used by people who manage and oversee projects, and possibly in my future career. Also, I hope to get a better understanding of how Drafter works and what details are needed in Construction Management.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to help construction managers keep track of their projects and progress. What are 3 common problems managers have when organizing and keeping track of these things, and what are 3 simple features a basic tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Problems
1. Difficulty Tracking Project Progress
2. Managing Budgets and Material Costs
3. Keeping Project Information Organized

Apps frequently used:
Procore.
Autodesk Construction Cloud.
Buildertrend.

AI Recommendation
Keep your website focused on these three main features:
Project Tracker: Create projects and track their completion.
Material Tracker: Add materials and calculate estimated expenses.
Task Manager: Create tasks, assign deadlines, and update their statuses.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think assigning deadlines is probably too complex for an introductory project. Can you make it more simple?
It changed assigning deadlines to Managing Multiple Projects At Once

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My idea has become more focused, because the LLM provided 3 clear features that the website should have that would make it most useful to solve real world problems.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I would first attend office hours, or meet with friends to see if they have ideas on what my problems are. If I use AI, it would be to help debug, not to write code for me.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If the LLM provides code that causes errors, I would first look through it myself, and compare it to example code in the python bakery textbook. If that doesn't help, I would attend office hours or post the code in discord to get help debugging.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I am a civil engineering major, what would be a good idea for a website built using Thonny and Drafter

Response:
Since you're a civil engineering major, I'd recommend building a website that solves a small, realistic civil engineering problem. That way, your project is not only creative but could also be something you use in future engineering classes.

Since you're using Thonny and Drafter, I'd focus on a project involving calculations, data visualization, and user input rather than something requiring a complicated database or backend.

Here are my top ideas.

5 civil engineering website ideas

1. Civil Engineering Project Calculator

A website where students can perform common introductory civil engineering calculations.

Possible features:

- *Calculate concrete volume for slabs, columns, and foundations.*
- *Calculate the area of a rectangular or triangular section.*
- *Estimate the amount of concrete needed for a project.*
- *Calculate basic unit conversions.*
- *Display calculation results and graphs.*

Example:

A user enters:

- *Slab length: 20 feet*
- *Slab width: 15 feet*
- *Slab thickness: 6 inches*

The website calculates:

- *Total volume: 150 cubic feet*
- *Concrete needed: approximately 5.56 cubic yards.*

Why it's good for Drafter:

- *Uses basic Python math.*
- *Requires user input.*
- *Can use multiple pages.*
- *Can display results in tables.*
- *Can include different calculation categories.*

You could even add a visual representation of the slab.

Important: This would be an educational calculator, not a tool for designing or approving real structures.

2. Road Design and Traffic Calculator

A website that helps users understand traffic patterns and basic roadway calculations.

Possible features:

- *Enter the number of cars passing through an intersection.*
- *Calculate average traffic volume.*
- *Compare traffic volume during different hours.*
- *Generate graphs showing traffic patterns.*
- *Estimate average vehicle flow.*

Example:

A user enters traffic counts:

Hour	***Vehicles***
8 AM	*120*
9 AM	*180*

10 AM 140

11 AM 90

The website generates a graph showing when traffic is busiest.

Drafter features: Matplotlib, Python lists, calculations, and user input.

3. Concrete Mix Estimator

A website that estimates the materials needed for a concrete project.

Possible features:

- *Enter the dimensions of a concrete slab.*
- *Calculate the required concrete volume.*
- *Estimate cement, sand, and aggregate quantities using a simplified assumed mix.*
- *Display material estimates.*
- *Show a breakdown of the quantities.*

This is particularly useful because it connects programming with construction materials.

Drafter features: Mathematical calculations, input fields, output tables, and possibly graphs.

Keep it to educational estimates using predefined mix assumptions rather than actual structural mix design.

4. Civil Engineering Site Explorer

A website that lets users explore construction locations and basic site information.

Possible features:

- *Display a map.*
- *Let users select or enter a location.*
- *Save notes about different construction sites.*
- *Record site observations.*
- *Display coordinates if geolocation is available.*

Drafter features: Maps, geolocation, local storage, and multiple pages.

This would probably have the most visually interesting interface, although it could be slightly more challenging.

5. Bridge Load Learning Simulator

A website that helps civil engineering students understand how loads affect simplified bridge or beam models.

Possible features:

- *Select a simplified beam type.*
- *Enter a load value.*
- *Calculate basic reactions for a simply supported beam.*
- *Display a diagram of the beam.*
- *Show how the reaction forces change when the load changes.*

Drafter features: Python calculations, conditional statements, input fields, and visualizations.

This is a little more challenging but would make an interesting engineering-focused project.

It should be presented as a simplified educational model, not a real bridge-design tool.

My comparison

Project	***Difficulty***	***Best feature***
Engineering Calculator	*Easy–Medium*	*Multiple calculations*
Traffic Calculator	*Easy–Medium*	*Graphs*
Concrete Estimator	*Easy–Medium*	*Material estimates*

Site Explorer	*Medium–Hard*	*Interactive map*
Bridge Simulator	*Medium–Hard*	*Engineering visualization*

What I would personally recommend

I'd choose Civil Engineering Project Calculator.

Here's why:

1. *It directly relates to your major.*
2. *You can build it with basic Python knowledge.*
3. *It demonstrates multiple programming concepts.*
4. *It can have a polished, professional-looking interface.*
5. *You could realistically use it in future engineering courses.*

I'd give it a name like CivilCalc: Your Civil Engineering Toolkit.

How it could fit your project assignment

Your uploaded project worksheet asks you to identify a problem, existing solutions, your solution, inputs, processing, outputs, and the core data your program tracks. Here's how the calculator idea could fit.

Assignment section	***Possible answer***
Problem	*Civil engineering students need a convenient way to perform basic engineering calculations.*
Prior solutions	*Online engineering calculators and unit conversion websites.*

Solution	*A centralized website for basic civil engineering calculations.*
Input	*Dimensions, measurements, and numerical values.*
Processing	*Python performs mathematical calculations and unit conversions.*
Output	*Calculated answers, tables, and visual representations.*
Data abstraction	*Calculation type, dimensions, units, and results.*

One question before narrowing down your project:

Which area of civil engineering interests you the most?

1. What area interests you most?

Structural engineering
Transportation / Highway engineering
Construction management
Environmental engineering
Geotechnical engineering
Water resources engineering
Not sure yet

2. What kind of website would you prefer?

Something practical I could use in future classes
Something visually impressive and interactive
Something involving lots of calculations
Something relatively easy to build but creative
Submit

I can then help you develop an idea that is interesting, realistic for a CS1 student, and strong enough for your final project.

I need to have multiple dataclasses, for loops, and a State dataclass. What would

incorporate those. DO NOT WRITE ANY CODE FOR ME

I would actually lean toward a Civil Engineering Project Manager or Construction Site Tracker. These ideas naturally incorporate multiple dataclasses, `for` loops, and a `State` dataclass without making the project unnecessarily complicated.

The key is to choose a project where you manage a collection of objects rather than just perform one calculation.

Here are four ideas that would work particularly well.

1. Construction Project Manager

Best overall fit
Project Management Software Tanzania | Tawala
TruGenie - AI Real Estate Intelligence Platform | TruBoard
Vergo AI | Free Construction Materials List Template
9

A website that allows users to create and manage small construction projects.

Possible dataclasses:

- `Project`*: Project name, location, budget, completion percentage.*
- `Task`*: Task name, start date, deadline, completion status.*
- `Material`*: Material name, quantity, price per unit.*
- `State`*: Stores the current project, list of projects, tasks, and materials.*

Where you would use for loops:

- *Loop through all projects to display them.*
- *Loop through tasks to identify unfinished tasks.*
- *Loop through materials to calculate total material costs.*
- *Loop through tasks to calculate project completion percentages.*

Example user experience:

A user creates a project called "Community Park Construction."

They add:

- *Site preparation*
- *Foundation installation*
- *Concrete pouring*

- *Landscaping*

They then add materials such as concrete, gravel, and lumber.

The website displays project progress, remaining tasks, and estimated material costs.

2. Civil Engineering Material Estimator

Taylos Trading | Premium Building Materials & Safety Tools in Doha, Qatar
Quantity takeoff is a challenging process, and construction professionals, especially quantity surveyors, know how demanding it can be. Even a minor error can lead to significant costs. However… | Saady Chohan
General Cost Estimator for Excel | General Cost Estimator
8

A website that estimates the materials and costs required for construction projects.

Possible dataclasses:

- `Material`*: Name, unit, price, quantity.*
- `Structure`*: Structure name, length, width, height.*
- `Estimate`*: Collection of materials and total estimated cost.*
- `State`*: Stores the structures and estimates created by the user.*

Where you would use for loops:

- *Loop through materials to calculate total costs.*
- *Loop through structures to calculate volumes.*
- *Loop through estimates to compare project costs.*
- *Loop through material lists to display a complete shopping list.*

Example user experience:

A user creates a concrete slab estimate.

They enter its dimensions and the website calculates its volume. They can then add multiple materials and see their estimated costs.

3. Road and Traffic Analysis Dashboard

AIカメラソリューションを活用した"交通量・通行量調査"「SCORER Traffic Counter Edge」をアップデートし、検知精度向上＆新機能を搭載！｜株式会社フューチャースタンダードのプレスリリース
North Dean Road & East University Drive Intersection Improvements - City of Auburn

PPT - Lec 5: Capacity and LOS (Ch. 2, p.74-88) PowerPoint Presentation - ID:6642260
8

A website that allows users to record traffic information and analyze traffic patterns.

Possible dataclasses:

- `Road`*: Road name, speed limit, number of lanes.*
- `TrafficCount`*: Road, time period, number of vehicles.*
- `Intersection`*: Name, connected roads, traffic counts.*
- `State`*: Stores all roads, intersections, and traffic records.*

Where you would use for loops:

- *Loop through traffic counts to calculate total vehicles.*
- *Loop through roads to display road information.*
- *Loop through intersections to identify the busiest intersection.*
- *Loop through traffic records to calculate average traffic volume.*

Example user experience:

A user records vehicle counts for three different intersections throughout the day.

The website generates graphs comparing traffic volumes and identifies which intersection had the highest recorded traffic.

4. Engineering Project Comparison Tool

ORIS - Materials Intelligence
Oracle Primavera What's New 2022
9

A website that allows users to compare different engineering projects based on cost, size, materials, and completion time.

Possible dataclasses:

- `Project`*: Project name, type, budget, duration.*
- `Material`*: Material name, quantity, cost.*
- `Metric`*: Measurement name, value, unit.*
- `State`*: Stores all projects being compared.*

Where you would use for loops:

- *Loop through projects to display comparisons.*

- *Loop through materials to calculate total costs.*
- *Loop through metrics to calculate averages.*
- *Loop through projects to identify the cheapest or largest project.*

How these ideas satisfy your programming requirements

Requirement	***How it gets incorporated***
Multiple dataclasses	*Different real-world objects with different properties*
For loops	*Processing collections of projects, materials, tasks, or measurements*
State dataclass	*Stores the application's current information*
User input	*Users create and modify records*
Data processing	*Calculations, comparisons, filtering*
Drafter interface	*Multiple pages, forms, tables, and graphs*

My recommendation

I'd rank them:

1. *Construction Project Manager – Best combination of functionality and manageable complexity.*
2. *Material Estimator – Best for mathematical calculations and civil engineering relevance.*

3. *Traffic Analysis Dashboard – Best for graphs and data analysis.*
4. *Project Comparison Tool – Best for organizing and comparing information.*

One important consideration: Since Drafter runs Python in the browser without a backend database, you can use local storage to preserve project information between visits.

I'd personally choose the Construction Project Manager. It gives you a natural reason to have four or more dataclasses, multiple collections to iterate through, and a meaningful State dataclass.

It also gives you plenty of opportunities to make the website look like a professional construction management application without requiring advanced programming.

I want to build a simple web tool to help construction managers keep track of their projects and progress. What are 3 common problems managers have when organizing and keeping track of these things, and what are 3 simple features a basic tool should have?"

For a construction project management website, three common problems managers face are:

1. Difficulty Tracking Project Progress

Construction projects involve many different tasks, workers, and deadlines. Managers can struggle to keep track of which tasks are completed, currently in progress, or behind schedule.

Simple feature: A project progress tracker that allows managers to create tasks, assign statuses, and view the overall completion percentage of a project.

2. Managing Budgets and Material Costs

Construction projects require many materials, and costs can add up quickly. Managers need to keep track of material quantities, prices, and total project expenses.

Simple feature: A material and budget tracker where managers can add materials, enter quantities and prices, and automatically calculate total costs.

3. Keeping Project Information Organized

Managers may oversee multiple projects at once. Keeping project details, deadlines, and notes organized can become difficult when information is scattered across different documents.

Simple feature: A project dashboard that stores project names, locations, budgets, deadlines, and important notes in one place.

Three existing tools to research

Some existing construction management platforms that address these problems include:

- *Procore.*
- *Autodesk Construction Cloud.*
- *Buildertrend.*

My recommendation for your project

Keep your website focused on these three main features:

1. *Project Tracker: Create projects and track their completion.*
2. *Material Tracker: Add materials and calculate estimated expenses.*
3. *Task Manager: Create tasks, assign deadlines, and update their statuses.*

These features would naturally allow you to incorporate multiple dataclasses (`Project`, `Task`, and `Material`) and a `State` dataclass that stores the application's information. You would also have plenty of opportunities to use `for` loops to process and display collections of projects, tasks, and materials.

This is a realistic scope for a three-week introductory programming project.

I think assigning deadlines is probably too complex for an introductory project. Can you make it more simple?

Absolutely. I would simplify the website to focus on organizing construction projects rather than managing schedules. You don't need deadlines, employee assignments, or complicated scheduling systems.

Here is a more manageable version:

Simplified Construction Project Manager

Problem 1: Difficulty Tracking Project Progress

Managers need to know which parts of a construction project have been completed and which still need work.

- Feature: A task tracker where users can add tasks and mark them as Not Started, In Progress, or Completed.

Problem 2: Keeping Track of Material Costs

Construction projects require different materials, and managers need to know how much they are spending.

- Feature: A material tracker where users enter material names, quantities, and prices. The website calculates the total estimated material cost.

Problem 3: Managing Multiple Projects

Managers may oversee several construction projects and need an easy way to organize their information.

- Feature: A project dashboard where users can create projects, view their details, and track their overall progress.

Simplified dataclasses

You could organize your program around these four dataclasses:

Dataclass	***Information stored***
`Project`	*Name, location, budget, tasks*
`Task`	*Task name, description, status*
`Material`	*Material name, quantity, price*
`State`	*Lists of projects, tasks, and materials*

How to incorporate your required programming concepts

Requirement	***Application***

Multiple dataclasses	*Project, Task, Material*
State dataclass	*Stores the application's information*
For loops	*Display projects, calculate material costs, count completed tasks*
If statements	*Determine whether tasks are completed*
Lists	*Store multiple projects, tasks, and materials*
Mathematical operations	*Calculate total material expenses and project progress*

Keep the project scope manageable

I would leave these features out:

- *Employee accounts and assignments.*
- *Calendar and deadline management.*
- *Automatic notifications.*
- *Real-time collaboration.*
- *Complicated budget forecasting.*

Your final website would essentially have three pages:

1. *Projects – Create and view construction projects.*
2. *Tasks – Add tasks and update their completion status.*
3. *Materials – Track materials and calculate their costs.*

This is a much more realistic project for a three-week CS1 course, while still giving you enough functionality to demonstrate your programming skills.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.